

A comparative evaluation of chain-of-thought-based prompt engineering techniques for medical question answering

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ABSTRACT

Background and objective: Large language models (LLMs) hold transformative potential for clinical decision-making in the rapidly advancing field of AI in medicine. This study evaluates how Chain-of-Thought (CoT) prompting techniques affect medical reasoning performance with consideration for clinical applicability.

Methods: Five LLMs (GPT-4o-mini, GPT-3.5-turbo, o1-mini, Gemini-1.5-Flash, Gemini-1.0-pro) were assessed via API access using CoT prompting methods exhibiting distinct cognitive characteristics. Evaluations included clinical datasets and non-clinical datasets. An iterative QA system was constructed to ensure consistent and reproducible results.

Results: o1-mini achieved the highest performance with 88.4 % accuracy in EHRNoteQA using MIMIC-IV discharge summaries. Traditional CoT consistently provided stable results across clinical datasets, while Interactive CoT showed variability - underperforming in complex medical reasoning tasks like MedMCQA (61.7 %) but performing well in clinical notes (83.5 % in EHRNoteQA). Statistical analysis revealed no significant differences between prompting methods across datasets ($p > .05$), though effect size analysis showed notable variations in method effectiveness.

Conclusion: Complex prompting techniques do not significantly enhance performance compared to simpler approaches. Dataset characteristics and model architecture have greater impact, suggesting simpler CoT methods may be more effective for clinical applications.

1. Introduction

Large language models (LLMs) have demonstrated notable capabilities across various natural language processing applications, including tasks like text generation, summarization, translation, code generation, and logical deduction [1–4]. In the medical sector, early initiatives have shown promising results, with LLMs capable of composing clinical documentation, successfully passing standardized medical examinations, and formulating answers to patient inquiries [5,6]. For a more seamless incorporation of LLMs into clinical practice, it is essential to understand their reasoning abilities in a clinical context [7]. In specialized fields like medicine, utilizing effective prompting techniques is vital to provide clear and safe guidance to the model [8].

Prompt engineering is crucial for optimizing LLMs, which involves refining prompts to improve LLM performance and communication clarity. It is essential for achieving optimal outcomes when working with LLMs [9]. The importance of prompt selection, especially in specialized

domains, cannot be overstated. Prompts act as vital inputs directing large language models to produce precise outputs, varying from straightforward phrases to more multimodal inputs like images [10]. The process of prompt engineering involves strategically crafting textual inputs to enhance the model's output accuracy and consistency significantly [11]. This process is essential for maximizing the potential of Generative AI models in various applications. Researchers and practitioners can improve the generated content's quality, relevance, and specificity by understanding and leveraging how these models interpret different prompts. This, in turn, expands the scope of Generative AI's applicability [12].

Chain-of-thought (CoT) prompting is an innovative technique designed to enhance the reasoning capabilities of large language models by steering them through a series of logical steps to solve complex problems [13]. OpenAI's o1 model now includes built-in chain-of-thought reasoning through reinforcement learning to enhance its reasoning capabilities [14]. It provides LLMs with exemplars that not

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only pose a question and its direct answer but also detail the intermediate reasoning steps that lead to that conclusion. This method effectively serves as a roadmap, guiding LLMs to mimic this thought process for unseen questions, thereby significantly improving their ability to tackle various reasoning tasks, from arithmetic to commonsense and symbolic reasoning challenges.

This paper investigates how different cognitive characteristics of Chain-of-Thought prompting methods affect the performance of large language models in medical question-answering tasks. While previous studies have explored LLMs’ potential in medicine, including clinical documentation and medical question-answering [5,6,15–18], our study extends this analysis by evaluating CoT-based prompting techniques across both clinical and non-clinical datasets. We specifically incorporated logical reasoning datasets to assess models’ ability to perform complex reasoning beyond simple pattern matching, addressing the limitations of clinical-only evaluations. Through API-based access to cost-effective LLMs, we evaluated our methods across diverse medical contexts while emphasizing logical reasoning capabilities, ensuring our findings are robust and generalizable for practical implementation in medical AI systems [19].

Our primary objectives are.

- To evaluate and compare the effectiveness of different CoT prompting methods in enhancing LLM performance in medical question-answering tasks, focusing on practical applications.
- To examine how these prompting methods perform across diverse medical contexts, from theoretical clinical knowledge to practical real-world data.
- To evaluate and identify effective prompting strategies that enhance LLM performance in clinical knowledge comprehension and complex medical reasoning tasks, thereby contributing to the standardization of prompt engineering practices in medical AI applications.

2. Methods

2.1. Prompt design

We designed several techniques based on the Chain-of-thought (CoT) approach: Traditional CoT [13,17], ReAct CoT [20], Interactive CoT, and Self-Consistency [21]. Traditional CoT induces a step-by-step approach to guide the model’s responses. ReAct CoT integrates reasoning and acting in a three-step cycle of thought (problem analysis), action (information retrieval or task execution), and observation (outcome evaluation), cycling through these steps until it finds the final answer. Interactive CoT was designed to examine the model’s meta-cognitive capabilities, focusing on its ability to monitor its reasoning process, track the evolution from initial thoughts to conclusions, and recognize uncertainties while requesting additional information when needed. Self-consistency CoT generates multiple reasoning paths and selects the most consistent answer through consensus, ensuring model outputs remain coherent and consistent across multiple attempts, enhancing reliability. While ReAct focuses on a single reasoning path with integrated actions, Self-Consistency emphasizes the convergence of

several reasoning trajectories. These CoT-based techniques were developed to improve model performance and create effective prompts tailored to domain-specific question and answering tasks (Table 1).

We collaborated with ChatGPT to develop prompt methods, which allowed us to create effective prompts while reflecting the unique characteristics of the medical field. For instance, to create the ReAct method, we prompted ChatGPT with: ‘Make various versions of the prompt templates to guide the model through the ReAct prompting (Wei et al., 2022) method to solve medical questions and answer tasks. Let’s think step by step.’ ChatGPT would generate several candidate templates following the ReAct structure when given this prompt. For example, a diagnostic reasoning template would be structured as: “Problem Statement” identifying the given symptoms, “Step-by-Step Reasoning” analyzing primary and secondary symptoms and considering potential causes, “Action” determining what additional information to request, and “Answer” suggesting the most likely diagnosis.

Another example would be a treatment-focused template that follows similar steps: stating the diagnosis, reasoning through treatment options while considering patient factors, recommending the most appropriate treatment, and justifying the final treatment plan. From these candidate templates, we selected and customized the most suitable one to align with our experimental requirements and ensure effective responses from the LLM. A basic prompt template was created to instruct the model and require a specific format (Table 2). This template corresponds to Control in this experiment to be compared with the different types of techniques, and it also works as a default prompt template.

Table 2
Prompt methods.

Methods	Brief Explanation	Prompt
Traditional CoT	Simplified chain of thought process	Let’s deduce the correct response step by step.
ReAct CoT	Cycle through reasoning and action until the final answer is reached	State your question and act to uncover information. Observe and analyze the outcome, then reflect on these insights step by step. Based on your reflection determine your next logical step.
Interactive CoT	Interactive update of thoughts to conclude	Identify the question and state your initial thought. Gather information related to it and update your thinking step by step. Keep interactive until you have a complete answer.
Self-Consistency CoT	Explore different paths to solve problems, reflect to identify the best answer and document the process for better decision-making	Define your question, search for diverse solutions. Explore different reasoning paths, reflect on outcomes, and document and analyze each reasoning step for self-consistency in decision making.

Table 1
A template of prompt engineering for the study.

Template
{Prompt Method} CAPITALIZED WORDS are placeholders. Fill in the placeholders with your rationale and answer (option). The answer must be a single letter (A,B,C,D, or E). Question: {question} Options: {options} Rationale: <RATIONALE> Answer: <ANSWER>

Note: The control technique was implemented without the {Prompt Method} part, while maintaining the same basic structure.

2.2. Dataset

This study evaluates three distinct categories of datasets: medical board exams, question answering using real-world clinical notes, non-clinical reasoning tasks, and multilingual tasks (Table 3), all structured in a multiple-choice question-answering format (Table 1). To ensure robustness and generalizability across diverse contexts, we selected datasets that represent a wide range of reasoning challenges. In the clinical domain, we utilize the MedQA dataset, which comprises standardized medical board examination questions from the United States, designed to assess physicians’ theoretical knowledge and clinical decision-making capabilities across various aspects of medical practice [22]. Additionally, MedMCQA incorporates clinical reasoning questions derived from medical entrance examinations, specifically targeting problem-solving abilities in medical contexts [23]. The clinical assessment is further enhanced by EHRNoteQA, which consists of over 960 questions based on MIMIC-IV discharge summaries, evaluating the comprehension and analysis of real-world clinical scenarios [24,25,26].

To ensure comprehensive linguistic coverage, we incorporate MMedBench, selecting Spanish and Chinese datasets due to their status as widely spoken languages. The Spanish dataset is derived from Head-QA [27], which consists of questions from Spain’s healthcare system professional entry exams, while the Chinese dataset is sourced from MedQA questions. These datasets, though different in origin and focus, provide a diverse representation of medical knowledge across languages [28].

To compare the effects of different prompting methods on nonmedical reasoning abilities, we employ GPQA, which consists of graduate-level questions across the domains of biology, physics, and chemistry [29]. This is complemented by BBH (Big Bench Hard), which presents challenging tasks focusing on multi-step reasoning and causal judgment [30]. These datasets were chosen to assess complex reasoning capabilities beyond medical contexts. The reasoning assessment is further strengthened by MMLU, which evaluates formal logic, critical thinking through logical fallacies, and general knowledge application [31]. Additionally, to manage computational resources efficiently, we randomly sampled 1000 entries from MedMCQA and 500 entries from MMedBench using a fixed random seed for reproducibility.

2.3. Model setting and automated QA framework

In our automated QA framework, we employed OpenAI and Google’s generative APIs to evaluate various prompting methods for GPT-4o-mini, GPT-3.5-turbo, o1-mini, Gemini-1.5-Flash, and Gemini-1.0-Pro. The temperature was set to 0 for all models, except the o1-mini, which has a fixed temperature of 1. Additionally, we implemented a 7-s delay between API calls to comply with rate limits. The framework processes questions iteratively through concurrent execution, handling each dataset and prompt method efficiently. The framework operates through a sequential data processing pipeline, selecting a prompt method, executing question-answering tasks, and generating the final output (Figs. 1 and 2).

Table 3
Dataset characteristics.

Category	Dataset	Size (k)	Domain
Clinical	MedQA	1.27k	Medical License
	MedMCQA	1.00k	Clinical Reasoning
	EHRNoteQA (MIMIC-IV)	0.96k	Real-world Clinical Documentation
Multilingual	MMedBench	0.50k	Spanish
		0.50k	Chinese
Non-clinical	GPQA	0.45k	Science
	MMLU	0.50k	Logic and Facts
	Big Bench Hard	0.18k	Causal Judgment

3. Result

To rigorously evaluate the effectiveness of prompt engineering techniques across diverse medical and reasoning tasks, we first compiled comprehensive performance metrics for each combination of models, datasets, and prompt methods (Table 4). These accuracy scores served as the foundation for our subsequent statistical analyses. We then implemented a comprehensive three-stage statistical framework. First, we conducted normality testing using the Shapiro-Wilk test to determine the appropriate statistical methods for each dataset. This guided our choice between parametric one-way ANOVA and non-parametric Kruskal-Wallis analyses. Second, we performed detailed performance analysis using bootstrap confidence intervals for robust estimation of performance metrics, implementing False Discovery Rate (FDR) correction to control for multiple comparisons while maintaining statistical power. Finally, we calculated Cohen’s d effect sizes to quantify the practical significance of performance differences between prompt engineering techniques relative to the Control method, with effect sizes categorized as small ($|d| \leq 0.5$), medium ($0.5 < |d| \leq 0.8$), or large ($|d| > 0.8$).

3.1. Overall performance metrics

Table 4 presents the accuracy scores across all experimental conditions, revealing notable performance variations. In clinical tasks, accuracy ranged from 58.14 % to 89.98 % (95 % CI [54.47, 91.83]), with EHRNoteQA showing consistently higher scores ($M = 89.98$ %, 95 % CI [88.13, 91.83]) across all models and prompt methods. MedQA ($M = 61.00$ %, 95 % CI [54.33, 67.67]) and MedMCQA ($M = 58.14$ %, 95 % CI [54.47, 61.81]) showed lower but stable performance. Non-clinical tasks demonstrated wider variability, with GPQA showing the lowest overall accuracy ($M = 39.72$ %, 95 % CI [35.46, 43.99]) while MMLU logical fallacies achieved notably higher scores ($M = 80.68$ %, 95 % CI [78.81, 82.55]). These baseline performance metrics provided the foundation for our subsequent statistical analyses examining the effects of different prompting techniques and model architectures.

3.2. Statistical framework and distribution analysis

The Shapiro-Wilk test revealed distinct distribution patterns across datasets. Clinical datasets predominantly exhibited normal distributions: MedQA ($W = 0.967, p = .125$), MedMCQA ($W = 0.972, p = .183$), and EHRNoteQA ($W = 0.981, p = .476$). For non-clinical tasks, MMLU-formal_logic ($W = 0.901, p < .001$), MMLU-logical_fallacies ($W = 0.913, p < .01$), and MMLU-global_facts ($W = 0.922, p < .01$) showed non-normal distributions. This distribution heterogeneity necessitated an adaptive statistical approach, employing ANOVA for normally distributed datasets and Kruskal-Wallis tests for non-normal distributions, with FDR correction for multiple comparisons.

3.3. Model performance analysis

Model performance analysis revealed significant variations across architectures. The o1-mini achieved the highest overall performance ($M = 73.93$ %, $SD = 13.53$ %, 95 % CI [70.05, 77.82]), followed by GPT-4o-mini ($M = 67.24$ %, $SD = 17.16$ %, 95 % CI [62.32, 72.17]), and Gemini-1.5-Flash ($M = 64.47$ %, $SD = 18.38$ %, 95 % CI [59.19, 69.74]). Lower performances were observed in GPT-3.5-turbo ($M = 57.20$ %, $SD = 19.09$ %, 95 % CI [51.72, 62.68]) and Gemini-1.0-pro ($M = 56.83$ %, $SD = 17.56$ %, 95 % CI [51.79, 61.87]). Compared to GPT-3.5-turbo, o1-mini demonstrated consistently large positive effects across all datasets ($d = 2.829$ to 15.071), while GPT-4o-mini showed substantial improvements ($d = 0.836$ to 15.087). Gemini-1.5-Flash maintained moderate to large positive effects ($d = 0.293$ to 7.498), whereas Gemini-1.0-pro often showed negative effects ($d = -1.293$ to -0.412) except in specific non-clinical tasks. These performance variations across models and datasets

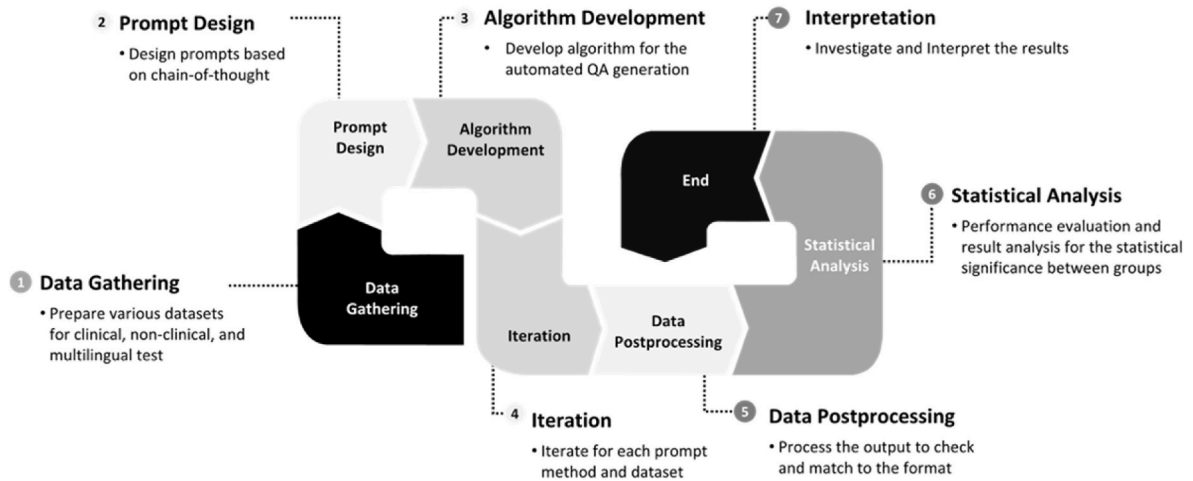


Fig. 1. Flow diagram of the study.

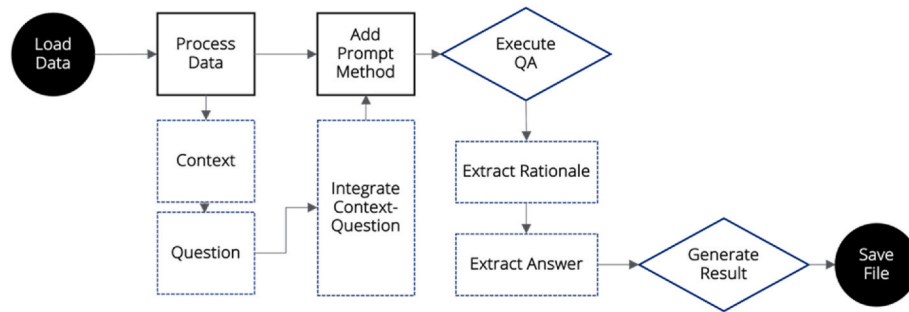


Fig. 2. Iterative QA evaluation framework.

Table 4
Comprehensive performance of LLMs and prompt methods.

Model	Prompt	MedQA	Med MCQA	EHR NoteQA	GPQA	BBH	MMLU (formal logic)	MMLU (logical fallacies)	MMLU (global facts)	MMedBench (Spanish)	MMedBench (Chinese)
GPT-4o-mini	TradCoT	70.50 %	68.00 %	92.93 %	37.30 %	64.70 %	42.10 %	82.20 %	60 %	89.60 %	69.40 %
	ReActCoT	56.20 %	66.10 %	93.14 %	39.30 %	68.40 %	41.30 %	84 %	62 %	86.90 %	70 %
	SelfConsistencyCoT	63.50 %	63.30 %	92.72 %	42.00 %	67.40 %	42.10 %	82.80 %	64 %	88.20 %	68.60 %
	InteractiveCoT	58.10 %	61.70 %	91.37 %	34.40 %	64.20 %	38.90 %	84 %	62 %	87.80 %	69.60 %
	Control	73.40 %	67 %	93.35 %	40.00 %	68.40 %	45.20 %	83.40 %	59 %	90.20 %	71.40 %
GPT-3.5-turbo	TradCoT	63.30 %	60 %	87.63 %	23.00 %	58.30 %	35.70 %	76.70 %	50 %	78.90 %	53.20 %
	ReActCoT	59.20 %	57.40 %	87.84 %	25 %	65.20 %	39.70 %	75.50 %	44 %	80.70 %	52.60 %
	SelfConsistencyCoT	60.60 %	56 %	87.42 %	25 %	59.90 %	36.50 %	75.50 %	32 %	79.10 %	53 %
	InteractiveCoT	27.70 %	39.20 %	87.32 %	24.00 %	59.40 %	38.10 %	66.90 %	45 %	77.90 %	55 %
	Control	61.30 %	59.30 %	87.53 %	27.00 %	54.50 %	42.10 %	78.50 %	54 %	81.70 %	54.80 %
o1-mini	TradCoT	85 %	68.40 %	91.16 %	52.70 %	67.90 %	61.10 %	84.70 %	64 %	93.20 %	79 %
	ReActCoT	83.10 %	68 %	89.71 %	49.10 %	67.40 %	55.60 %	84 %	61 %	92.60 %	77.80 %
	SelfConsistencyCoT	85 %	69.30 %	89.40 %	52 %	68.40 %	59.50 %	83.40 %	57 %	93.40 %	75.20 %
	InteractiveCoT	84.10 %	69.20 %	91.37 %	48.90 %	67.40 %	64.30 %	82.80 %	64 %	93 %	75.40 %
	Control	85.30 %	70.30 %	92.10 %	54.50 %	63.60 %	61.90 %	86.50 %	61 %	93.80 %	79.20 %
Gemini-1.5-Flash	TradCoT	61.60 %	53 %	97.40 %	52.20 %	68.50 %	44.40 %	82.20 %	50 %	88.80 %	71.10 %
	ReActCoT	61.40 %	50.20 %	97.61 %	36.80 %	57.80 %	47.60 %	85.30 %	44 %	87.60 %	75.10 %
	SelfConsistencyCoT	60 %	52.20 %	97.92 %	45.50 %	65.80 %	41.30 %	84 %	47 %	88.80 %	61.80 %
	InteractiveCoT	44.50 %	49.10 %	79.63 %	26.60 %	58.80 %	47.60 %	82.20 %	52 %	89.80 %	59.20 %
	Control	62 %	59 %	93.56 %	52 %	71.10 %	44.40 %	83.40 %	48 %	88.60 %	74.90 %
Gemini-1.0-pro	TradCoT	52.90 %	50.80 %	83.26 %	45.30 %	59.90 %	39.70 %	76.10 %	52 %	78.70 %	62.20 %
	ReActCoT	47.10 %	45.40 %	86.90 %	30.80 %	54.50 %	36.50 %	75.50 %	45 %	77.70 %	30.50 %
	SelfConsistencyCoT	51.60 %	47.20 %	86.90 %	39.70 %	62.60 %	38.10 %	82.20 %	40 %	77.90 %	61.80 %
	InteractiveCoT	31.80 %	48.40 %	87.63 %	44.20 %	59.90 %	37.30 %	79.10 %	51 %	78.30 %	59.20 %
	Control	35.80 %	55 %	83.68 %	45.80 %	56.10 %	37.30 %	76.10 %	49 %	76.70 %	30.30 %

are visualized in Fig. 3, highlighting the consistent superiority of o1-mini and the varying effectiveness of different models across task categories.

3.4. Dataset-specific performance

In clinical tasks, EHRNoteQA demonstrated superior performance

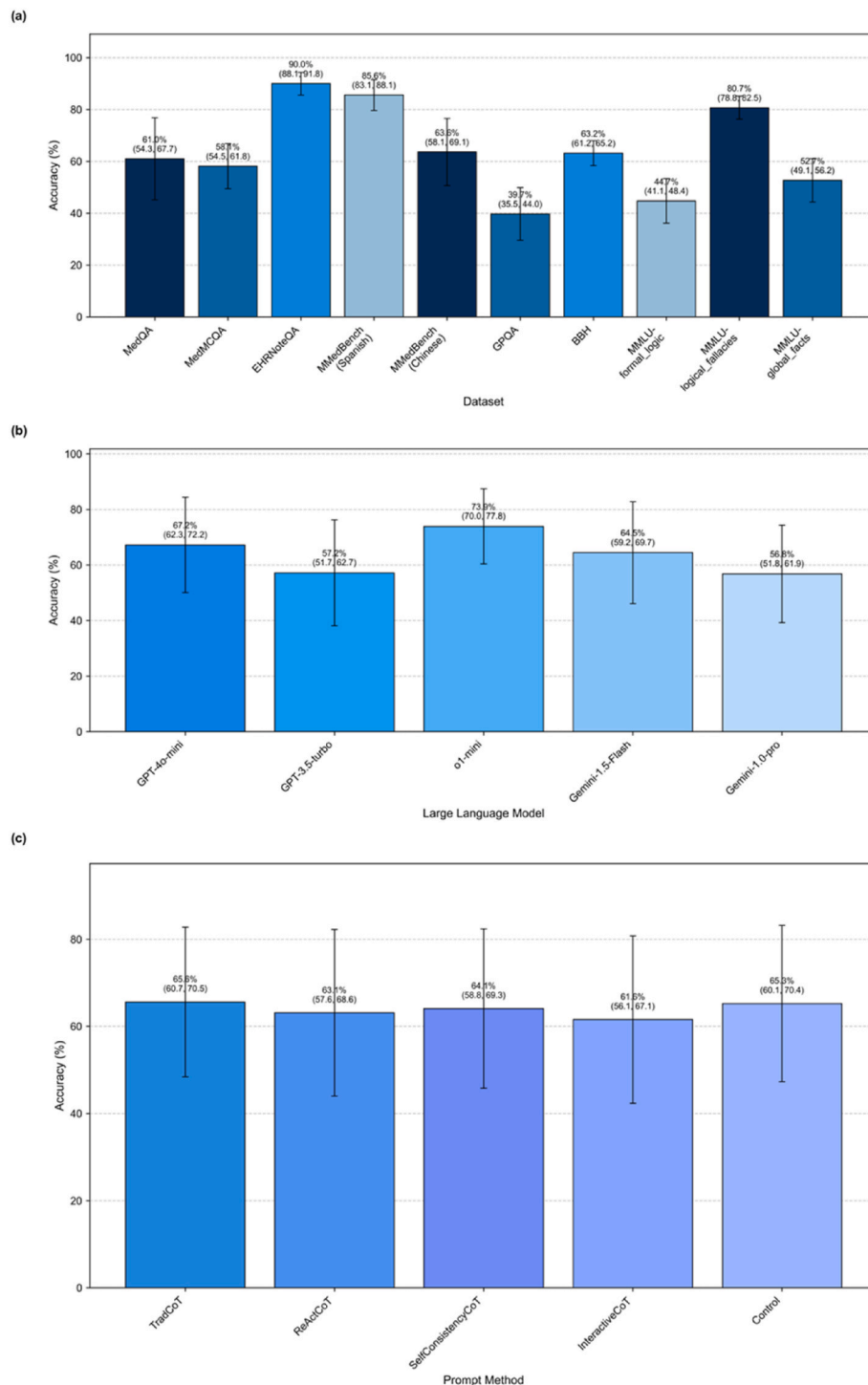


Fig. 3. Performance comparison of different datasets, models, and prompts.

($M = 89.98\%$, $SD = 4.40$, 95 % CI [88.13, 91.83]), significantly outperforming both MedQA ($M = 61.00\%$, $SD = 15.84$, 95 % CI [54.33, 67.67]) and MedMCQA ($M = 58.14\%$, $SD = 8.70$, 95 % CI [54.47, 61.81]). In non-clinical tasks, MMLU logical fallacies achieved the highest performance ($M = 80.68\%$, $SD = 4.43$, 95 % CI [78.81, 82.55]), while GPQA showed lower performance ($M = 39.72\%$, $SD = 10.12$, 95 % CI [35.46, 43.99]). In multilingual tasks, Spanish language tasks ($M = 85.60\%$, $SD = 5.93$, 95 % CI [83.10, 88.09]) performed better than Chinese tasks ($M = 63.61\%$, $SD = 12.98$, 95 % CI [58.14, 69.08]).

3.5. Prompt engineering effectiveness and effect sizes

In our analysis, Control serves as a baseline prompt method that provides basic instructions without any specific reasoning technique. Each CoT method's effectiveness is measured through Cohen's d effect sizes calculated against this Control condition, enabling standardized comparison across different datasets. Statistical analysis revealed no significant differences between prompt engineering techniques after FDR correction across datasets (MedQA: $F = 0.875, p = .496$; MedMCQA: $F = 0.618, p = .655$; EHRNoteQA: $F = 0.482, p = .748$). Traditional CoT demonstrated the most consistent performance ($M = 65.61\%$, $SD =$

17.17, 95 % CI [60.68, 70.54]), while Interactive CoT showed the lowest overall performance ($M = 61.59\%$, $SD = 19.21$, 95 % CI [56.07, 67.10]).

Effect size analysis revealed varying impacts across different contexts: Traditional CoT maintained stable performance compared to Control ($d = -0.301$ to 0.308 , lowest in MMLU logical fallacies and highest in Chinese tasks). Interactive CoT showed consistent negative effects ($d = -0.905$ to -0.566 , most pronounced in MedMCQA) and ReAct CoT demonstrated variable effects ranging from medium negative to small positive ($d = -0.760$ to 0.229 , lowest in GPQA and highest in EHRNoteQA). SelfConsistency CoT showed moderate variability ($d = -0.623$ to 0.362 , lowest in MMLU global facts and highest in BBH causal judgement) These effect sizes indicate performance deviations from the Control method, with positive values suggesting improvement and

negative values indicating performance decline relative to the basic prompting approach. The distribution of performance across different prompt engineering techniques is illustrated in Fig. 4, demonstrating the relative stability of Traditional CoT compared to other methods.

4. Discussion

This research evaluates the effectiveness of various Chain-of-Thought (CoT) prompting methods in enhancing large language models' performance across clinical and non-clinical reasoning tasks. Our study found that prompting strategies did not yield statistically significant performance differences across datasets. Instead, model architecture emerged as a key determinant, with o1-mini achieving the

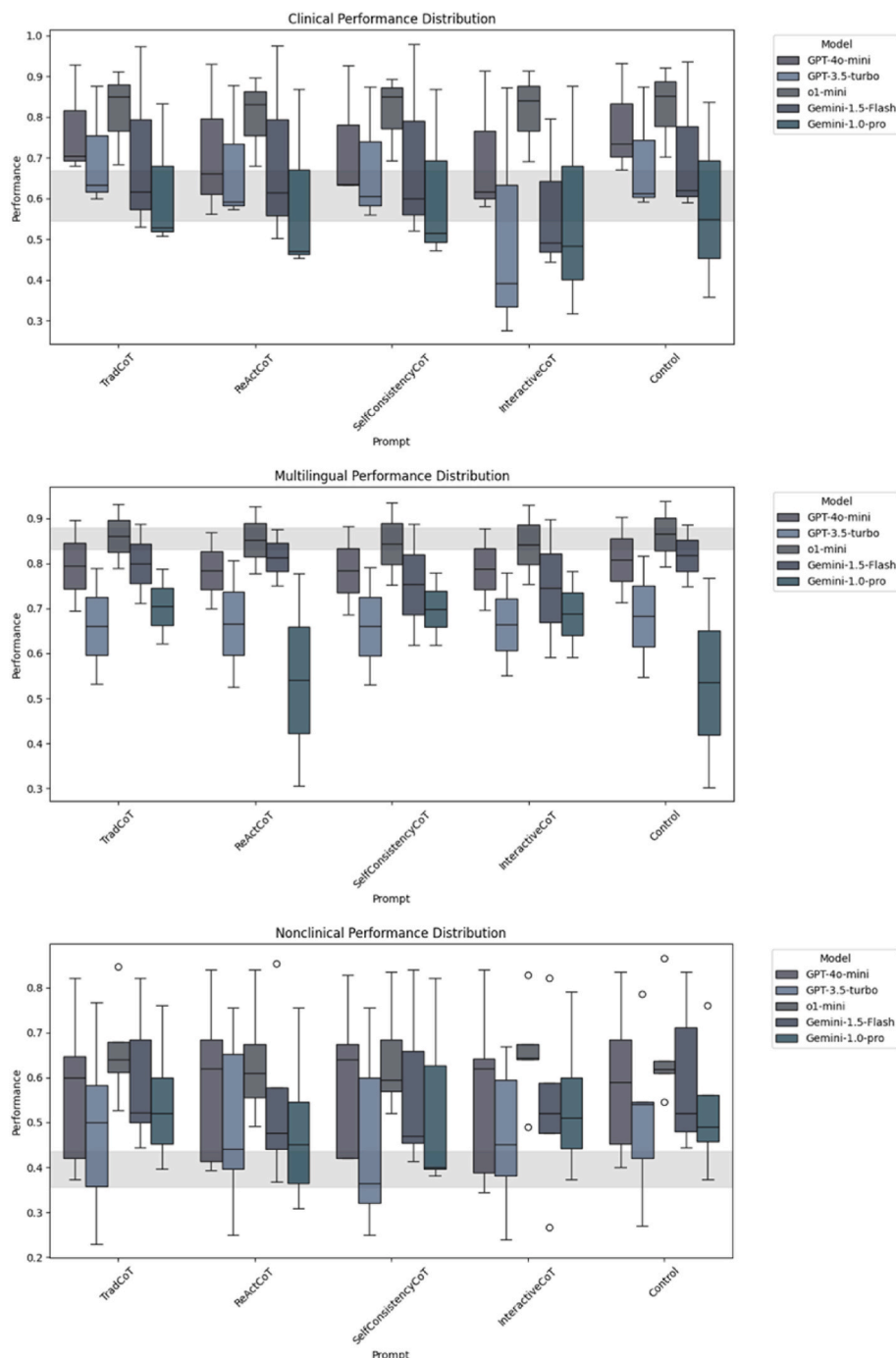


Fig. 4. Distribution of performance across prompt methods.

highest overall performance, followed by GPT-4o-mini and Gemini-1.5-Flash. The performance patterns across different task types demonstrate that base model capabilities and dataset characteristics, rather than prompting techniques, are the crucial factors in determining effectiveness. This suggests that for real-world clinical applications, the focus should be on selecting appropriate models rather than utilizing complex prompting strategies. Interactive CoT consistently showed negative effects across medical knowledge tasks and real-world datasets, suggesting that complex interactive approaches may not be optimal for clinical applications. In contrast, ReAct CoT demonstrated variable performance between question-answering linked to discharge summaries and scientific reasoning tasks. These differences suggest that the complexity introduced by advanced prompting methods may not always translate into better outcomes for certain task types, particularly in medical contexts where clarity and reliability are crucial.

Traditional CoT maintained stable performance across datasets by following a straightforward sequential reasoning path. In contrast, Interactive CoT's attempt to leverage metacognitive abilities - monitoring reasoning processes, tracking evolution of thoughts, and recognizing uncertainties - often resulted in lower performance, particularly in complex reasoning tasks. This was evident in medical knowledge tasks, where Interactive CoT showed considerable performance drops in particularly MedMCQA. Similarly, ReAct CoT's integration of reasoning and acting in a three-step cycle showed variable performance across different tasks, suggesting that the additional complexity of action steps may interfere with the core reasoning process. These patterns indicate that while Traditional CoT provides a stable foundation through its simple, direct reasoning approach, both Interactive CoT and ReAct CoT's more sophisticated mechanisms may introduce unnecessary complexity that interferes with the model's reasoning process, particularly in tasks requiring sophisticated medical knowledge and scientific reasoning.

Our results demonstrate that while there were no statistically significant differences between prompting methods, prompt engineering strategies affect performance differently based on their complexity level, with simpler approaches often yielding more reliable outcomes across different clinical reasoning tasks. This insight is particularly relevant for applications such as clinical decision support systems, medical education, and clinical documentation [32,33–36].

Previous research has shown that generalist language models like GPT-4 and GPT-3.5 can perform well in answering medical questions without extensive fine-tuning [15,16]. However, the challenge of interpretability in applying large language models in medicine, mainly due to their black-box nature, has been a significant barrier. Savage et al. (2024) addressed this challenge by examining LLMs' ability to simulate clinical reasoning in diagnostic cases [17]. Their study revealed that GPT-4 could successfully replicate physicians' diagnostic reasoning processes while maintaining diagnostic accuracy through specialized diagnostic reasoning prompts. This approach presents a substantial leap towards integrating LLMs into clinical settings by providing an interpretable rationale for their recommendations, aiding physicians in assessing the trustworthiness of LLM responses for patient care.

Another study by Wang et al. (2024) examined the effectiveness of prompt engineering strategies in medical applications [18]. Their systematic evaluation of LLMs' consistency with established medical guidelines for treating osteoarthritis revealed that specific prompting methods could significantly improve the accuracy and reliability of AI responses. Their findings particularly emphasized how different prompting strategies could affect the model's adherence to evidence-based guidelines, demonstrating the critical role of appropriate prompt engineering in enhancing LLMs' performance for medical applications. While Savage et al. (2024) and Wang et al. (2024) suggested that specialized prompt strategies could enhance LLM performance in medical domains, our findings demonstrate that increased prompt complexity does not necessarily guarantee better performance. Despite the lack of statistical significance after FDR correction and the

consistent performance of Traditional CoT suggest that simpler, more straightforward reasoning approaches may be more effective in medical question-answering.

A particularly notable finding is the impact of dataset characteristics on model performance. The high performance on EHRNoteQA demonstrates the potential of LLMs in real clinical data applications, while the lower performance on GPQA reveals limitations in general scientific reasoning tasks. The large negative effect size observed with Interactive CoT in MedMCQA suggests that complex cognitive approaches may actually interfere with core reasoning processes. These findings emphasize that in developing clinical decision support systems, considering the model's fundamental capabilities and dataset characteristics may be more crucial than optimizing prompting techniques.

The performance variations among different models provide additional insights into the relationship between model architecture and medical reasoning capabilities. The o1-mini achieved the highest overall performance, significantly outperforming other models including GPT-4o-mini, particularly excelling in medical licensing exams and maintaining high performance in multilingual contexts (Spanish and Chinese). This robust performance may be attributed to its reinforcement learning-based chain-of-thought capabilities, as o1-mini is specifically designed to think through problems by producing a long chain of thought before providing answers [14]. In contrast, GPT-4o-mini and GPT-3.5-turbo showed more variable performance patterns. While GPT-4o-mini demonstrated strong capabilities in complex reasoning tasks through its end-to-end multimodal training [37], GPT-3.5-turbo's performance fluctuated across different medical context.

These differences suggest that model training strategies play crucial roles in determining performance across various medical reasoning tasks, independent of prompting methods. These model-specific differences were not limited to OpenAI models. Similar variations were observed in the Gemini models. Gemini 1.5 Flash, despite being designed for efficiency, demonstrates improved performance over Gemini 1.0 Pro across many benchmarks, including reasoning and knowledge tasks. This is likely due to Gemini 1.5 Flash's use of parallel computation of attention and feedforward components, as well as online distillation from the larger Gemini 1.5 Pro model [38,39]. Additionally, Gemini 1.5 Flash is trained with higher-order preconditioned methods for improved quality. While these advancements show promise for medical AI applications, specific comparisons in medical reasoning tasks between these models require further investigation.

Recent research by Goh et al. (2024) and Chen et al. (2024) complements our findings, suggesting that the integration of LLMs into clinical practice requires more nuanced approaches than simply enhancing model performance [32,40]. Based on our experimental results, we propose the following concrete implementation strategies for clinical settings:

First, when developing clinical decision support systems, the focus should be on selecting models with strong baseline capabilities rather than implementing complex prompting techniques. Our findings with o1-mini demonstrate that models trained specifically for sequential reasoning perform more reliably in medical contexts. Second, the integration of LLMs should be tailored to specific clinical tasks. For instance, EHRNoteQA results suggest that these systems could be particularly effective for clinical documentation and discharge summary analysis, while more complex diagnostic reasoning tasks may require different approaches. Lastly, the implementation should prioritize transparency and interpretability. Traditional CoT's consistent performance suggests that straightforward reasoning paths may be more practical for healthcare professionals to verify and integrate into their workflow.

These implementation strategies should be validated through pilot studies in real clinical settings. Future research should focus on how LLMs can effectively synergize with healthcare professionals in real clinical settings. Rather than concentrating solely on model performance, it is crucial to develop implementation strategies that explore the practical integration of these models alongside human medical

expertise. This approach will help ensure that AI systems complement and enhance clinical decision-making processes while maintaining the essential role of healthcare professionals.

5. Conclusion

This research examines the effectiveness of different Chain-of-Thought prompting methods in enhancing LLMs' medical reasoning capabilities. Our findings reveal that prompting strategies did not yield statistically significant performance differences across datasets. Instead, model architecture and training strategies emerged as key determinants of performance, with o1-mini's reinforcement learning-based chain-of-thought capabilities demonstrating superior capabilities across diverse medical contexts. Traditional CoT's stable performance, particularly in clinical datasets linked to real-world clinical notes, suggests that simpler prompting approaches may be more reliable for clinical applications.

However, our study has several limitations. First, while our experiments demonstrate the effectiveness of simpler prompting approaches, they were conducted in controlled settings that may not fully reflect the complexities of real clinical environments. Second, the performance variations across different medical contexts suggest that model capabilities and dataset characteristics play more crucial roles than prompting strategies. This highlights the need for multidimensional evaluation frameworks that consider both technical performance metrics and practical clinical implementation challenges.

Based on these findings, we propose that future research should focus on two key areas. First, investigating how model selection and concise prompting approaches can be optimized for specific clinical tasks, rather than pursuing complex prompting strategies. Second, developing implementation strategies that effectively integrate LLMs into clinical workflows while maintaining the essential role of healthcare professionals. This approach will help ensure that LLM-based AI systems serve as practical and collaborative tools that create synergistic relationship between humans and AI, leading to enhanced clinical decision-making.

CRedit authorship contribution statement

Sohyeon Jeon: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Hong-Gee Kim:** Supervision.

Ethics statement

This research did not involve any human subjects or animal experiments. All data processing and analysis were conducted in accordance with ethical research practices and data protection guidelines.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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